

Welcome and General Information

B.Sc. in Computer Science, Fifth Semester

Autumn 2026

University of Southern Denmark

Courses

- DM546 Compiler construction
- DM571 Software engineering
- DM585 Distributed and web programming
- DM586 Networks and cybersecurity

 This semester has several projects.

Pizza Meeting

- Information about elective courses offered in Spring 2027
- Information about bachelor projects for Spring 2027
- Usually in mid-November
- More information on itsLearning at 'Students at IMADA'

Welcome and General Information

DM585 Distributed and Web Programming

Marco Peressotti

Autumn 2026

University of Southern Denmark

- Marco Peressotti

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- Start the subject with '[DM585]' (I get a lot of emails)
 - Avoid messages on itsLearning (I do not get notifications)

- 🏢 Office hours: open door (best effort) or book a meeting

- 📍 Office V25-412b-1 (map [↗](#))



- Viktor Strate Kløvedal (TA)

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- 🏢 Office hours: book a meeting

✓ DM574 Introduction to Programming

- ⋮ Structured programming, programming by contracts, ...

✓ DM575 Object-Oriented Programming

- ⋮ Management of complexity in software design and development.

- ⋮ Modelling application domains using OO design patterns.

✓ DM580 Functional Programming

- ⋮ algorithmic modularity, abstraction

✓ DM584 Concurrent Programming

- ⋮ Multitasking, asynchrony, scaling on the same (multicore) machine ...

DM585 Distributed and Web Programming

- ⋮ Scaling across machines and organisations, web technologies, ...



Why?

- The Web is a feature of our lives, it is critical infrastructure for our societies and economies.
- The Web is by design an heterogeneous ecosystem
 - It relies on a several technologies
HTTP, HTML, CCS, JS, ...
 - It is distributed and decentralised
How do we scale? How do we handle failures? how do we handle consistency? ...
- Approaching the development of distributed web applications might be frustrating
Where do I start? Which frameworks do I pick? ...



How?

- **Distributed Web Applications (5 ECTS)**

We will follow a practical, project-based approach:

- We will build a web application and scale it up by distributing it
- We will see few but fundamental principles
- We will see several technologies (some together, some on your own)
- We will write a lot of code

- **Innovation (2.5 ECTS)**

You will play a startup, create a business plan, and develop a product.

This is joint with AI510, it is ok to work across degrees.

The distributed web application project can be your prototype.

Learning Outcomes

- Explain fundamental problems, techniques, and solutions for distributed web applications;
- Identify and apply common design patterns;
- Design, implement, and test a distributed web application model for concrete scenarios;
- Realise innovative information processing based on a business model.

Content

Programming internet-based distributed systems

- Web architecture and common web technologies
- Microservice architectures and API design patterns
- Replication, consistency, and fault tolerance.
- Consensus and agreement in distributed systems.

Element A (5.0 ECTS): A long group project + oral defence

- Groups of three (recommendation: same as innovation project)
- Runs from week 43 to week 51
- Two parts: (1) build a web app and (2) make it scale
- You can hand in (1) and (2) separately or as one.
- Oral defence by group in week 3 (to be confirmed)

Element B (2.5 ECTS): Innovation Project (joint with AI510)

- Groups of three (two if a group of three is not available)
- Runs from week 40 to week 48
- Mid-way review (i.e., a short presentation) in weeks 43–45

Sep	36
	37
	38
	39
Oct	40
	41
	42
	43
	44
Nov	45
	46
	47
	48
Dec	49
	50
	51

Week		Element A				Element B		
Sep	36	Lectures	Web Foundations					
	37							
	38							
39					Project	Introduction		
Oct	40		Web Applications					
	41							
	43			Project		Part 1	Mid-way reviews	
	44							
Nov	45		Distributed Web Applications					
	46							
	47							
48						Hand in		
Dec	49							
	50							
	51					Hand in		

Use it but own your work

- It can automate many of the programming tasks you will encounter.
- Own your solutions: you are expected to understand, explain, and defend everything you hand in, architecture, algorithms, and implementation.
- Beware the illusion of understanding: being able to follow AI-generated code does not mean you understand it.
- A useful strategy: practise a concept or technique yourself until confident, then use AI to automate its application.

Exams

- AI is permitted in supporting roles and must be clearly disclosed.
- You are responsible for everything you hand in.

Slides, Exercises, Tutorials

- Your notes
- Course slides/code and weekly exercise sheets (partially covered during exercise classes)
- Online tutorials, tools documentation and community

Some interesting books (optional)

- *Building Microservices (2nd Edition)* by Sam Newmann.
- *Microservices Patterns* by Chris Richardson.
- *Flask Web Development* by Miguel Grinberg.

Announcements and communication

- Check the folder *Course General Information* on itsLearning
- All important information will be posted/announced on itsLearning